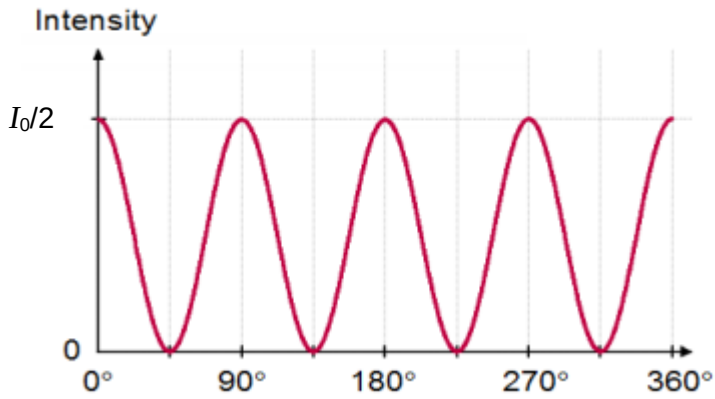


9	(a)	(i)	Wavelength $\lambda = 2.4 \text{ cm}$ Path difference $\Delta x = 0.4 \text{ cm}$ $\frac{\Delta x}{\lambda} \times 2\pi = \frac{0.4}{2.4} 2\pi = \frac{\pi}{3} = 1.05 \text{ rad}$ The waves are out of phase with ... phase difference = 60° or $\pi/3 \text{ rad}$ Unit : degree or rad depending on the value stated for phase difference	1 1
		(ii)	From Fig. 9.1, we can see that <u>any point on one wave has a constant separation with a point of the same phase on the other wave</u>, hence the two waves are always in constant phase difference or have a constant <u>phase relation</u>. Therefore, they are said to be coherent.	1 1
	(b)	(i)	$I_0/2$ [When an unpolarised light passes through a polariser of any axis, the intensity will be halved]	1
		(ii)	$I_0/2 \cos^2(\phi)$	1

	(iii)	 1m for correct shape 1m for correct angle for each cycle 1m for correct max intensity value	
(c)	(i)	$\sin \theta_1 = \lambda/d$ $= (590 \times 10^{-9})/(0.2 \times 10^{-3}) = 2.95 \times 10^{-3}$ $\theta_1 = 2.95 \times 10^{-3} \text{ rad}$	1 1
	(ii)	$\theta_2 = 3 \times 2.95 \times 10^{-3} = 8.85 \times 10^{-3} \text{ rad}$	1
	(iii)	$\text{Width} = 2 \times 2.95 \times 10^{-3} \times 0.75$ $= 4.4 \text{ mm}$	1 1
	(iv)	For patterns to be just resolved, central maximum of one beam must lie on the first minimum of the other (Rayleigh's criteria) $\text{angle} = 2.95 \times 10^{-3} \text{ rad}$	1 1
(d)	(i)	$x = \frac{\lambda D}{a}$ $\lambda = \frac{ax}{D}$ $= \frac{(2.5 \times 10^{-3})(0.45 \times 10^{-3})}{1.83}$ $= 6.15 \times 10^{-7} \text{ m} = 615 \text{ nm}$	1 1
	(ii)	Advantage: More light is able to pass through and thus the interference fringes will be brighter (more contrasting), and more easily observed. Disadvantage: The light passing through each slit diffracts (spread out) less, hence the interference pattern formed will cover a smaller area on the screen. In other words, there will be fewer interference fringes that can be observed.	1 1

10	(a)	(i)	The rate of decay is not dependant on physical conditions like	1
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